

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Even Semester)

Innovative Teaching Method Execution

UNIT IV Noise Characterization

Degree, Semester& Branch: IV Semester B.E.ECE B

Course Code & Title: EC8491 Communication Theory

Name of the Faculty member: Mr.K.Ragavan

Name of the Topic: Shot Noise

Name of the Innovative Practice: Multimedia clip

Date & Duration: 05.03.2020 & 10 Minutes

Description of Multimedia clip:

Multimedia clip is a brilliant and innovative new way to encourage students to communicate stories, ideas and concepts in a creative and original way. It can be particularly useful as a tool to encourage the creativity of students. Animated videos are a one stop solution that can grab the attention of the students. With its entertaining visuals, it can clearly comprehend the concept. Since it is an analogy to cartoons, students will get hooked to it. By using animation videos, students learn analytical thinking as to how to present a scenario.

Goals (Learning Outcomes):

- ❖ The students will be able to analyse the noise performance in communication systems.

Use of appropriate methods:

Justification for choosing the topic using multimedia clip:

The basic of communication starts from modulation and demodulation process and transmitted through wired channel. One must know about how the noises can affect the system and channel. So I have chosen this topic as a multimedia clip for better understanding and how the signal is affected by the shot noise.

Effective presentation

Details of the Implementation:

The multimedia clip explaining about the noise and its types especially shot noise affected the communication system was played using the LCD projection available in the classroom.

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ❖ The assessment of effectiveness of the activity was felt when asked questions in the classroom after showing the video.

- ❖ Due to this activity, the students were able to recall the topics they had learnt in the classroom and they were also able to reproduce it in the exam.

Reflective Critique:

- **Benefits of conducting this activity:**

- ❖ This video gives conceptual explanation about the communication system affected by noise
- ❖ The students showed more interest by listening to multimedia clip, apart from theory concept.
- ❖ The students gave oral feedback that now they got a clear idea about the detection of amplitude modulated wave and they requested to play more such videos like this

- **Challenges:**

Though it was a multimedia clip, certain explanations were given by me in between the multimedia clip for better understanding of the students. So it took some 2 to 3 minutes extra time than what I planned to complete.

Samples images of the activity:



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1	2			2		2		2

CO1: The students will be able to analyse the noise performance of AM and FM systems.

References:

1. J.G.Proakis, M.Salehi, "Fundamentals of Communication Systems", Pearson Education 2014
2. Simon Haykin, -"Communication Systems", 4th Edition, Wiley, 2014
3. <https://www.youtube.com/watch?v=GdunxCyMGUw>